

Supervised Machine Learning with KNIME

The KNIME (Konstanz Information Miner) analytics platform is perhaps the strongest and most comprehensive free platform for drag-and-drop analytics, machine learning, statistics, and ETL (extract, transform and load) models. This article will help develop a supervised machine learning decision tree algorithm through the KNIME tool for a car evaluation data set.



Data science, data pre-processing, modelling, analysis, and visualisation are all enabled within KNIME, the Konstanz Information Miner. The workflows can run both through the interactive interface and also in batch mode in this free and open source software. These two setups allow for easy local job management and regular

process execution. KNIME integrates various components for machine learning and data mining through its modular data pipelining ‘Lego of Analytics’ concept.

Let’s develop a supervised machine learning decision tree algorithm through the KNIME tool for the car evaluation data set. Decision trees are created for the car model based on the technical features, pricing criteria (overall cost, maintenance cost), comfort, size of luggage boot, safety, etc. The decision tree learner node and the predictor nodes are connected with both trained data and test data for training and evaluating the model with better output. Then the decision tree predictor is configured with the scorer node to generate the required evaluation performance metrics and to report data for user interface visualisation.

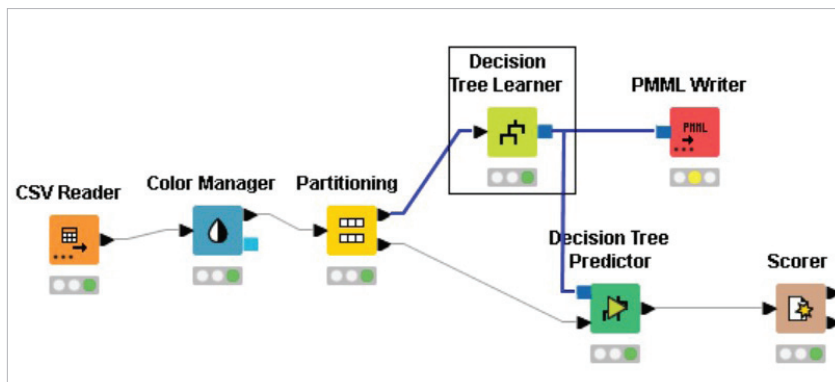


Figure 1: Car evaluation decision tree model KNIME workflow

List of nodes

CSV Reader Node	CSV Reader
Color Manager Node	Color Manager
Partitioning Node	Partitioning
Decision Tree Learner Node	Decision Tree Learner
PMML Writer Node	PMML Writer
Decision Tree Predictor Node	Decision Tree Predictor
Scorer Node	Scorer